# Mini Project 3 Report

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## 1 Introduction

Accurate moisture determination in biomass materials is crucial for optimizing energy production, improving storage conditions, and ensuring efficient processes. Moisture content significantly impacts the calorific value, handling properties, and overall usability of biomass in industrial applications. Traditional methods for moisture measurement, such as oven-drying, are time-consuming and labor-intensive, making them impractical for real-time monitoring and large-scale operations.

This report focuses on developing a machine learning model to predict biomass moisture content using Near-Infrared (NIR) spectroscopy. The dataset consists of spectral data collected from various solid biomass samples, including pine and spruce wood chips, bark, forest residues, and sawdust, along with reference moisture values obtained through standard laboratory methods.

Different preprocessing techniques, such as Savitzky-Golay smoothing, Standard Normal Variate (SNV), and Multiplicative Scatter Correction (MSC), are applied to enhance signal quality and reduce noise. Machine learning models, including Partial Least Squares (PLS), Support Vector Regression (SVR), and Artificial Neural Networks (ANN), are explored to determine the most accurate and reliable approach for moisture prediction.

By leveraging machine learning and advanced spectral preprocessing techniques, this project aims to provide a fast, reliable, and non-destructive method for biomass moisture determination. The findings can contribute to improved efficiency in biomass-based energy production and support the development of real-time monitoring solutions in industrial applications.

## 2 Data Analysis

The dataset used in this study consists of Near-Infrared (NIR) spectral measurements collected from various biomass samples, including pine and spruce wood chips, bark, forest residues, and sawdust. Each sample is characterized by its spectral absorbance values over a specific wavenumber range, along with reference moisture content values determined through standard laboratory methods. The spectral data provides a rich source of information for predicting moisture content using machine learning techniques.

Table 1 summarizes the key characteristics of the dataset.

Table 1: Summary of spectral dataset characteristics.

| **Parameter** | **Value** |
|---|---|
| Recorded spectral range | 834-2500 nm (12000 – 4000 cm$^{-1}$) |
| Number of measurements per sample | 5-7 |
| Spectral resolution | 16 cm$^{-1}$ |
| Number of data points in each spectrum | 1037 |
| Number of tested samples | 125 |

Each sample's spectral data consists of 1037 absorbance values corresponding to different wavenumbers, providing detailed insights into the material composition. The spectral resolution of 16 cm$^{-1}$ ensures high precision in capturing variations in moisture-related spectral features. Since multiple measurements are taken per sample (5-7 replicates), this helps reduce variability and improve model reliability.

Each sample is characterized by:

- **Spectral Absorbance:** A vector containing 1037 absorbance values measured at specific wavenumbers within the range of 12000–4000 cm$^{-1}$ (834–2500 nm). This high-dimensional data represents the unique spectral fingerprint of each sample.
- **Reference Moisture Content:** A numerical target variable representing the actual moisture percentage in the biomass sample, determined through laboratory analysis.
- **Sample ID:** A unique identifier for each biomass sample.
- **unnamed 1:** Most likely a leftover from conversion of different file formats.

The feature vectors in this dataset are composed of continuous absorbance values across multiple wavelengths. Each sample is represented as:

$$X = [A_{\lambda_1}, A_{\lambda_2}, ..., A_{\lambda_{1037}}]$$

where $A_\lambda$ denotes the absorbance value at wavelength $\lambda$. These values are continuous positive numbers.

There is **773** samples in total. All of them have a float64 type.

Several challenges arise when analyzing this dataset:

- **High Dimensionality:** The large number of spectral features (1037 per sample) increases computational complexity and may lead to overfitting in machine learning models.
- **Multicollinearity:** Adjacent wavenumbers often exhibit high correlation, which can negatively impact model interpretability and redundancy in feature representation.
- **Spectral Noise:** Measurement noise, baseline shifts, and instrument variations can affect the quality of spectral data. Preprocessing techniques such as Savitzky-Golay smoothing and Standard Normal Variate (SNV) transformation help mitigate these effects.
- **Data Sparsity and Limited Samples:** With only 125 tested samples, generalization to unseen data may be challenging, necessitating careful cross-validation and robust model selection techniques.
- **Outliers and Variability:** Variations in sample preparation, measurement conditions, or biomass heterogeneity can introduce inconsistencies in spectral responses.

Addressing these issues through proper data preprocessing, feature selection, and model regularization is crucial for developing an accurate and robust predictive model for biomass moisture determination.

By leveraging this dataset, machine learning models can be trained to establish a relationship between the spectral signatures and moisture content, allowing for accurate predictions without the need for traditional time-consuming laboratory techniques.

# 3 Pre-processing Techniques

Column *Unnamed: 1* was dropped due to it most likely being a leftover column from conversion to different file formats.

Firstly the standard procedure was followed: detecting null values, duplicates, and outliers. There were **0 null values**, **0 duplicates**, and **0 outliers**. To identify potential outliers in the spectral dataset, the **Interquartile Range (IQR)** method is employed. The IQR method is a widely used technique for detecting extreme values that deviate significantly from the main distribution of data.

For each spectral wavelength (feature) in the dataset, outliers are detected using the following steps:

1. Compute the first quartile ($Q_1$) and third quartile ($Q_3$) of the absorbance values for a given wavelength.
2. Determine the Interquartile Range (IQR), defined as:

$$IQR = Q_3 - Q_1$$

3. Establish the lower and upper bounds for detecting outliers:

$$\text{Lower Bound} = Q_1 - 1.5 \times IQR$$

$$\text{Upper Bound} = Q_3 + 1.5 \times IQR$$

Any data point falling below the lower bound or above the upper bound is considered an outlier.

The outlier detection process is applied iteratively across all spectral features ($n$ wavelengths). Given a dataset with spectral measurements recorded at multiple wavelengths, the algorithm systematically computes outlier thresholds for each column.

Four random rows were chosen and their boxplots are presented in the figure 1. It is visible here as well that there are no outliers (no dots at the end of the range).

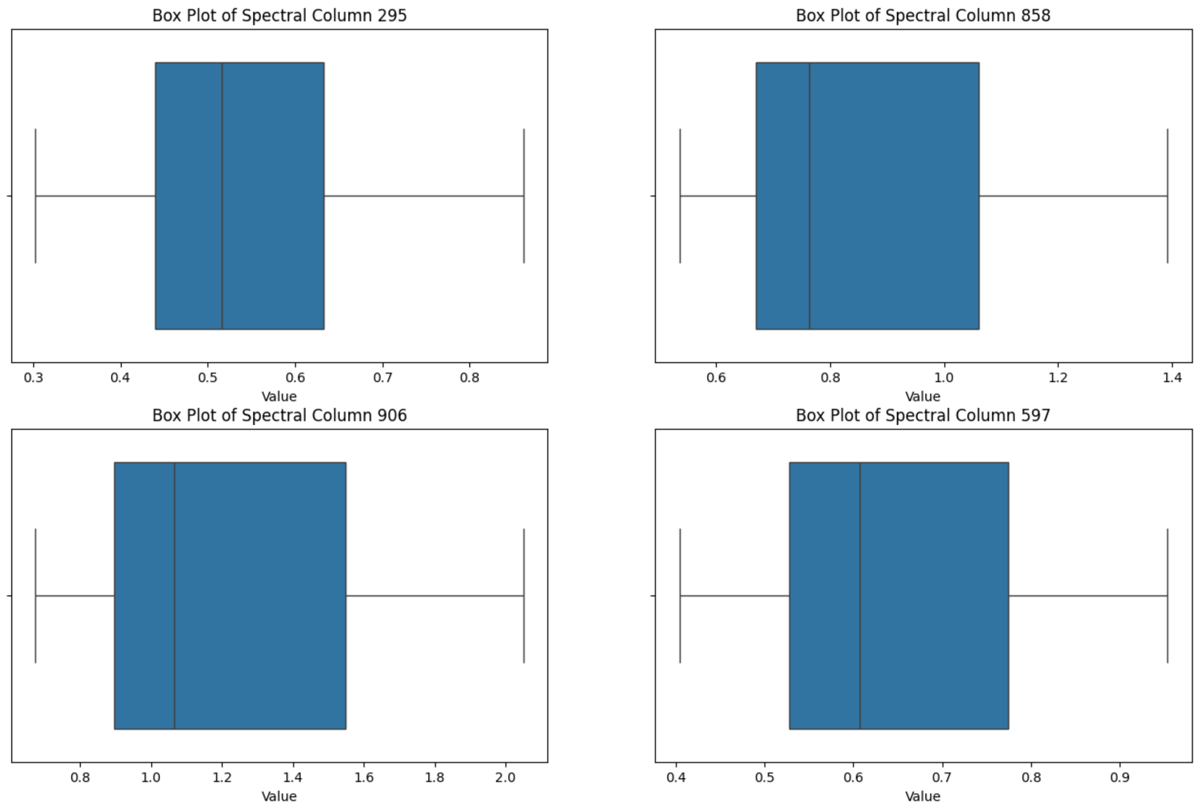


Figure 1: *Boxplots for four randomly chosen rows*

Four random rows were chosen and their histograms are presented in the figure 2.

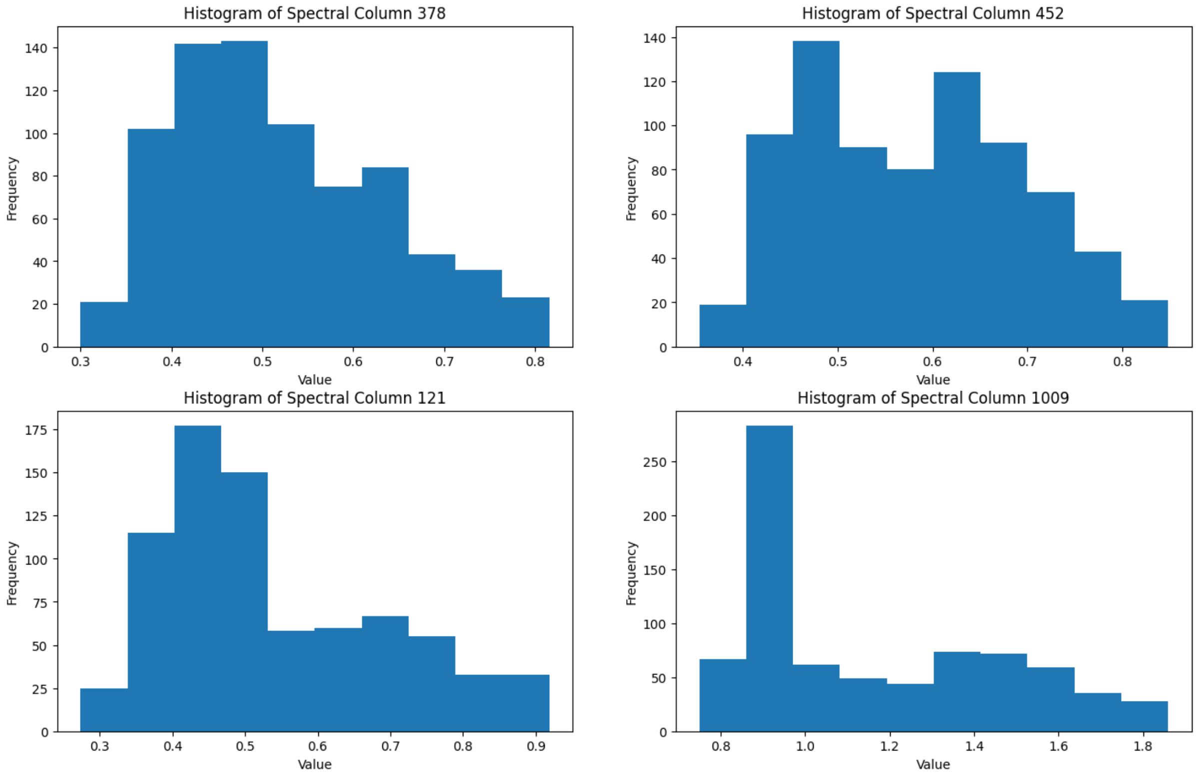


Figure 2: *Histograms for four randomly chosen rows*

Six random rows were chosen and their spectra are presented in the figure 3.

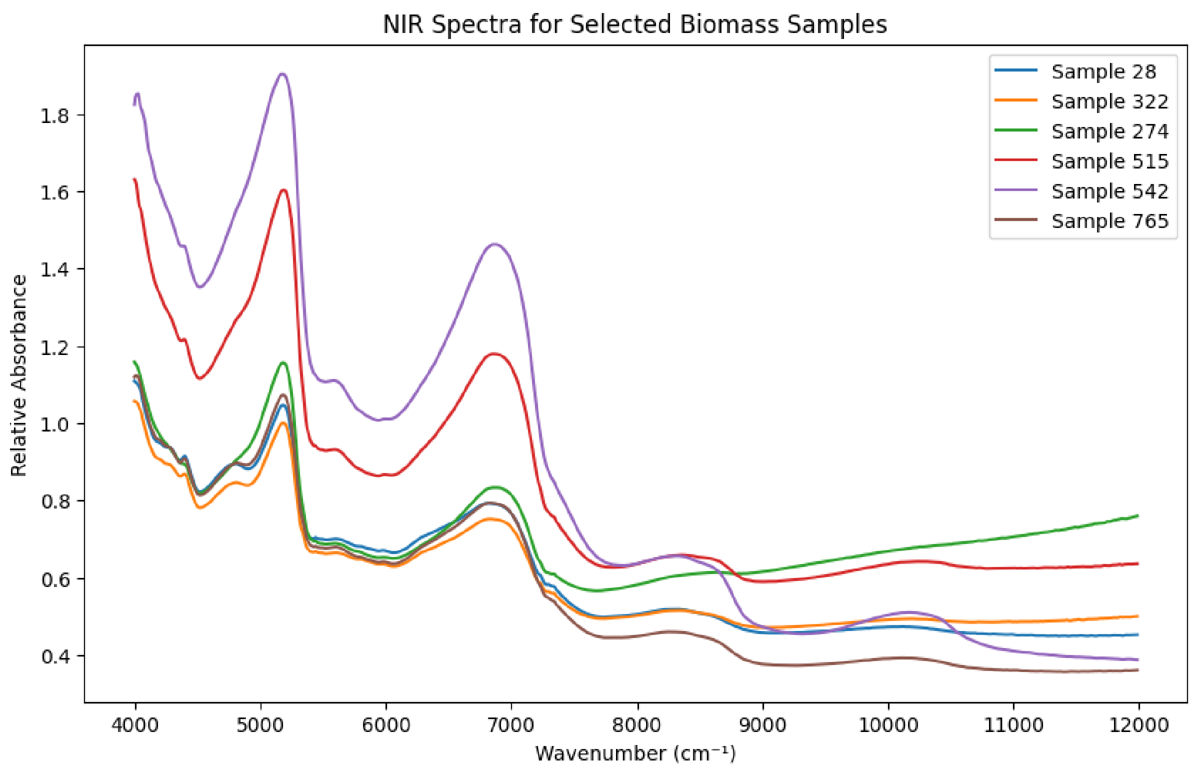


Figure 3: *NIR spectra for six randomly chosen rows*

Correlation was also explored, and below 5 most correlated features are listed, along with their correlation values with the target variable *moisture*. Even though correlation is high, that is expected due to the continuous nature of the spectra. This is known as collinearity and is common in spectroscopic data.

| **Feature** | **Correlation Value** |
|---|---|
| 5114.791 | 0.899023 |
| 5122.505 | 0.899021 |
| 5107.076 | 0.898987 |
| 5130.220 | 0.898978 |
| 5099.361 | 0.898928 |

Table 2: Features with High Correlation Values

Three different advanced preprocessing techniques, which are used in the chemometric approaches, were tested.

## 3.1 Savitzky-Golay Smoothing

Savitzky-Golay (SG) smoothing is a digital filter that reduces noise in spectral data while preserving essential features such as peaks and edges. This method fits a polynomial of order $p$ within a moving window of size $w$, and replaces each data point with the polynomial-smoothed value. The general formulation for the smoothed signal $\tilde{x}_i$ is:

$$\tilde{x}_i = \sum_{j=-k}^{k} c_j x_{i+j} \tag{1}$$

where $c_j$ are the polynomial coefficients, and $k = \frac{w-1}{2}$. SG smoothing is particularly useful for improving signal quality without significantly distorting the spectral shape. Additionally, SG filtering can be used to compute derivatives of spectral data, which helps to enhance spectral features and remove baseline shifts. The first derivative is computed as:

$$\frac{dx_i}{d\lambda} = \sum_{j=-k}^{k} c'_j x_{i+j} \tag{2}$$

where $c'_j$ are the derivative coefficients derived from the polynomial fit. The first derivative enhances peak separation and removes constant background shifts, while the second derivative further suppresses broad background variations and highlights fine spectral details.

## 3.2 Standard Normal Variate (SNV)

SNV is a normalization technique used to correct for scatter effects in spectral data by standardizing each spectrum individually. Given a spectral vector $\mathbf{x} = [x_1, x_2, ..., x_n]$, SNV transformation is performed as:

$$x'_i = \frac{x_i - \mu}{\sigma} \tag{3}$$

where $\mu$ and $\sigma$ are the mean and standard deviation of the spectral vector, respectively. This transformation centers the data around zero and scales it to unit variance, effectively removing baseline shifts and intensity variations.

## 3.3 Multiplicative Scatter Correction (MSC)

MSC is another method for compensating for spectral variations caused by light scattering. It assumes that each spectrum $\mathbf{x}$ is a linear transformation of an ideal reference spectrum $\mathbf{r}$:

$$x_i = a + br_i + \epsilon_i \tag{4}$$

where $a$ and $b$ are the offset and scaling coefficients, and $\epsilon_i$ represents noise. The MSC transformation estimates these parameters and applies a correction:

$$x_i' = \frac{x_i - \hat{a}}{\hat{b}} \tag{5}$$

where $\hat{a}$ and $\hat{b}$ are computed by regressing each spectrum against the mean reference spectrum. MSC effectively reduces additive and multiplicative scatter effects, enhancing spectral consistency.

One random sample is taken and all of these transformations are applied to the original spectrum. Transformations include: SG, SG 1st derivative, SNV, and MSC. Figure of the result can be seen in 4. It can be seen that SG spectrum looks exactly the same as the original spectrum.

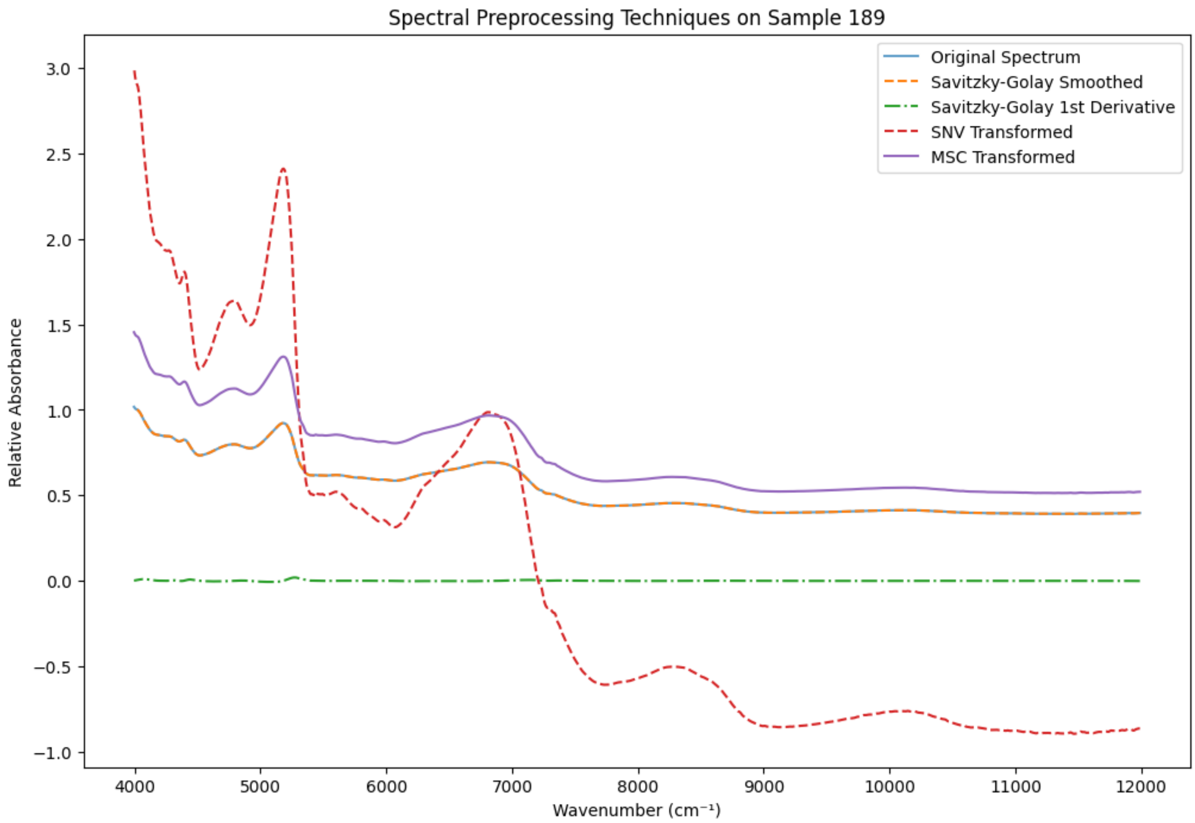


Figure 4: *Spectral preprocessing techniques on a random sample*

# 4 Modeling Approaches

Three models were chosen for this project: Partial Least Squares (PLS), Support Vector Regressor (SVR), and Artificial Neural Network (ANN). All of these are proven to be effective for complex regression problems.

## 4.1 Partial Least Squares Regression (PLS)

Partial Least Squares (PLS) regression is a linear modeling approach that combines features of Principal Component Analysis (PCA) and multiple linear regression. It is particularly effective when dealing with high-dimensional, collinear spectral data.

**Characteristics:**

- PLS reduces the dimensionality of the predictor variables by extracting latent components that maximize the covariance between predictors and the target variable.
- Unlike PCA, which selects components based solely on variance, PLS components are chosen to maximize predictive power.

- PLS is robust against multicollinearity, making it well-suited for spectral data where neighboring wavelengths are often highly correlated (as can be seen from the table 2).

**Usage Scenarios:**

- PLS is widely used in chemometrics and spectroscopy for predicting chemical properties from spectral measurements.
- It is suitable when the number of predictors far exceeds the number of observations.
- Effective for datasets with multicollinearity issues, where standard linear regression would fail.

**Benefits:**

- Handles high-dimensional data efficiently.
- Reduces overfitting by selecting only the most informative components.
- Works well even when predictors exhibit strong collinearity.

## 4.2 Support Vector Regression (SVR)

Support Vector Regression (SVR) is a non-linear regression technique based on Support Vector Machines (SVMs), designed to model complex relationships between features and target values.

**Characteristics:**

- SVR aims to find a function that has a maximum margin from the training data while minimizing prediction errors within a predefined tolerance $\epsilon$.
- Uses kernel functions (e.g., linear, polynomial, radial basis function) to map input features into a higher-dimensional space for capturing non-linear patterns.
- The regularization parameter $C$ controls the trade-off between model complexity and prediction accuracy.

**Usage Scenarios:**

- SVR is useful when the relationship between features and target values is non-linear.
- Applied in spectroscopy when complex interactions exist between wavelengths and moisture content.
- Effective for small to medium-sized datasets where feature selection is difficult.

**Benefits:**

- Can model both linear and non-linear relationships using different kernel functions.
- Robust to outliers due to the $\epsilon$-insensitive loss function.
- Provides good generalization even with limited data.

## 4.3 Artificial Neural Networks (ANN)

Artificial Neural Networks (ANNs) are deep learning models inspired by the human brain, capable of capturing highly non-linear relationships in data.

**Characteristics:**

- Consists of multiple layers of neurons, including input, hidden, and output layers.
- Each neuron applies an activation function (e.g., ReLU, sigmoid) to introduce non-linearity.
- Optimized using backpropagation and gradient descent algorithms.

**Usage Scenarios:**

- ANN is effective when dealing with large, complex datasets with non-linear patterns.

- Suitable for spectroscopy applications where intricate relationships exist between spectral features and target variables.
- Used when other models fail to capture underlying dependencies in the data.

**Benefits:**

- Highly flexible and capable of learning complex, non-linear relationships.
- Can automatically extract relevant features from high-dimensional data.
- Scalable to large datasets with sufficient computational resources.

The data was split into training and testing groups, with 80% being used for training, and 20% being used for testing. That gave **618** samples for training, and **155** samples for testing. This was used only for grid search and for obtaining predictions for scatter plot of results. To get the actual results for the models, cross-validation with 5 folds was employed using the whole dataset.

Models were compared based on the preprocessing technique used, $RMSE$, which is the Root Mean Squared Error metric and $R^2$, which is coefficient of determination.

$R^2$ (Eq. 6) metric measures how well the regression model fits the data. It represents the proportion of the variance in the dependent variable that is predictable from the independent variable(s).

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{6}$$

- $y_i$ is the actual value of the target variable.
- $\hat{y}_i$ is the predicted value of the target variable.
- $\bar{y}$ is the mean of the actual values.
- $n$ is the number of observations.

The numerator represents the sum of squared residuals (prediction errors), and the denominator represents the total sum of squares (variance of the actual values). An $R^2$ value closer to 1 indicates a better fit.

$RMSE$ (Eq. 7) measures the square root of the average squared difference between the actual and predicted values. It provides an interpretable metric in the same unit as the target variable, making it easier to assess the magnitude of prediction errors.

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{7}$$

- $y_i$ is the actual value of the target variable.
- $\hat{y}_i$ is the predicted value of the target variable.
- $n$ is the number of observations.

Lower RMSE values indicate better predictive performance, as they imply that the predicted values are closer to the actual values while maintaining interpretability in the original unit of measurement.

# 5 Results

Grid search is commented out in the notebook because it is slow, but the results it give were used as hyperparameters. It is important to note that ANN uses some randomness to initialize weights, which is why results can differ across runs. Results presented here are the ones I got when running.

- PLS was used with $n\_components = 20$
- SVR was used with $C = 200$ and $epsilon = 0.1$

- ANN was used with $max\_iter = 1000$, $alpha = 0.001$, $hiddem\_layer\_sizes = (100, 50)$, and $learning\_rate\_init = 0.001$
- SG filter was used with $window\_length = 11$, $polyorder = 2$, and $deriv = 0$

Full results overview sorted by $R^2$ is given in the table 3.

Table 3: Model Performance with Different Preprocessing Techniques

| **Model** | **Preprocessing** | $R^2$ | **RMSE** |
|---|---|---|---|
| PLS | SNV | 0.921643 | 2.937461 |
| PLS | MSC | 0.912723 | 3.103111 |
| PLS | Savitzky-Golay | 0.907576 | 3.278609 |
| SVR | MSC | 0.880301 | 3.841875 |
| SVR | SNV | 0.879272 | 3.841742 |
| ANN | SNV | 0.866512 | 4.372196 |
| ANN | MSC | 0.790197 | 5.099448 |
| SVR | Savitzky-Golay | 0.787986 | 5.113743 |
| ANN | Savitzky-Golay | 0.657109 | 6.151653 |

As it can be seen, PLS model is the best performing one, regardless of the preprocessing method used. When comparing the preprocessing techniques, SNV and MSC are competing for the best one. SG is consistently performing the worst across three models.

## 5.1 Visualization

Scatter plot was used to visualize performance of the models on the test set. X-axis represents actual values, while the y-axis represents predicted values. Red dotted line is the perfect performance.

Best performing model for PLS was with SVN preprocessing. Results can be seen in the figure 5.

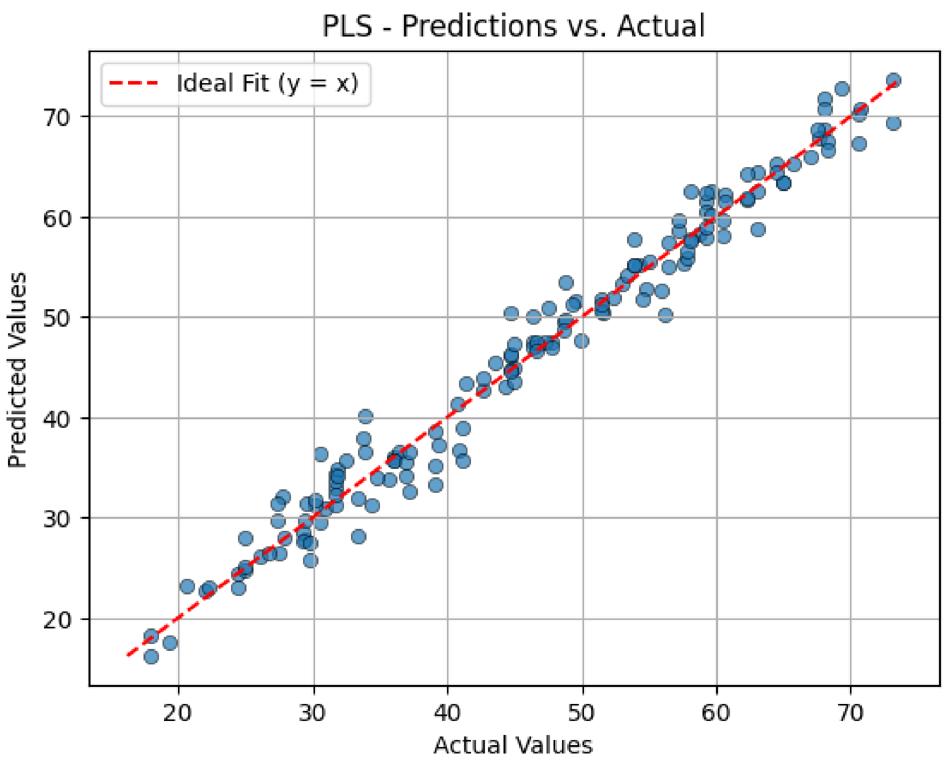


Figure 5: *Scatter plot visualization for PLS with SVN preprocessing results*

Best performing model for SVR was with MSC preprocessing. Results can be seen in the figure 6.

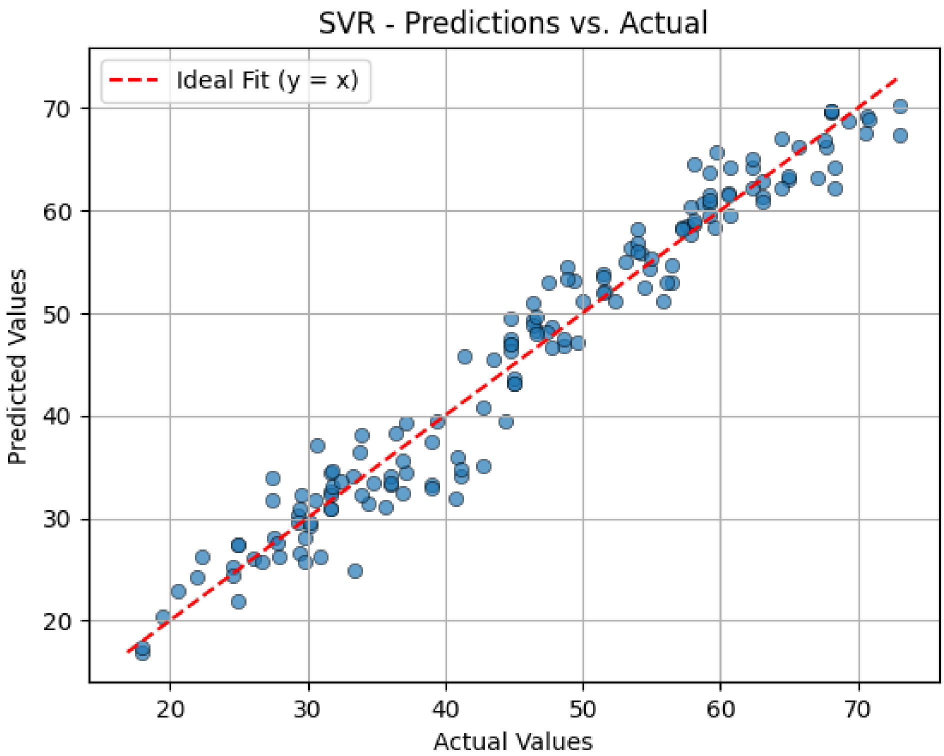


Figure 6: *Scatter plot visualization for SVR with MSC preprocessing results*

Best performing model for ANN was with SVN preprocessing. Results can be seen in the figure 7.

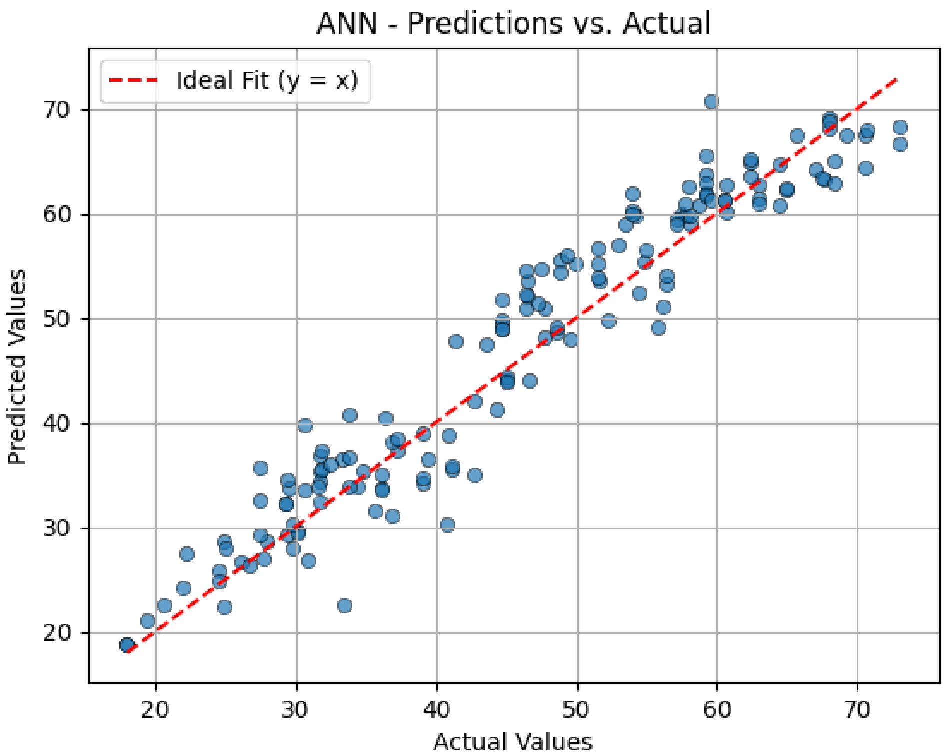


Figure 7: *Scatter plot visualization for ANN with SVN preprocessing results*

# 6 Conclusion

As it can be seen from the table 3, the $R^2$ score ranges from 0.65 to 0.92, which means that some models are much better suited for this task, and that the choice of preprocessing technique impacts the result greatly. It can also be observed that PLS outperforms the other two modes. Below are given some of the possible reasons as to why that is.

- *Handling High-Dimensional and Correlated Data*: The dataset consists of spectral data, where features (wavelengths) exhibit high collinearity. PLS is designed to effectively manage multicollinearity by projecting the original data onto a new set of orthogonal components while maintaining strong relationships with the target variable (moisture content). In contrast, SVR and ANN may struggle with highly correlated features, leading to suboptimal generalization.

- *Feature Extraction Ability*: Unlike standard regression models, PLS extracts latent variables that capture the most variance relevant to the target variable. This enables the model to reduce noise and focus on meaningful features for prediction. SVR and ANN do not inherently perform this type of feature extraction, making them more dependent on preprocessing techniques and feature selection.

- *Sample Size Considerations*: Given a limited dataset, ANN typically requires more data to generalize well, as deep learning models are prone to overfitting when trained on small datasets. SVR, while effective in many regression tasks, is sensitive to feature scaling and kernel selection, making it trickier to optimize compared to PLS.

- *Stability Across Preprocessing Methods*: PLS maintains high $R^2$ scores across different preprocessing techniques, demonstrating its robustness to transformations such as Standard Normal Variate (SNV), Multiplicative Scatter Correction (MSC), and Savitzky-Golay filtering. SVR and ANN exhibit greater performance variations, indicating their increased sensitivity to preprocessing choices.

In general, SNV and MSC preprocessings work the best. For PLS SNV is the best performing one, while for SVR MSC is the best performing one. For ANN it differs between runs, so it cannot be said for certain. The consistently worst performing preprocessing technique is SG.

ANNs require more data to be trained, which is why data augmentation would be beneficial. Additionally, dimensionality of the data is quite high, which is why using a dimensionality reduction technique, such as the Principal Component Analysis, before training could be beneficial for models that struggle with high complexity. Furthermore, these preprocessing techniques can be combined to achieve even better results.

In conclusion, the greatest challenge here was the domain knowledge. Knowing the data allows us to make informed decisions while choosing the preprocessing techniques and the models, and without this knowledge, we are just trying things out blindly. It was nice that we got to try it out, but a lot more research is needed to use this in practice.